From Paws to People:

A Statistical Assessment of Health Trajectories in Common Dog Breeds

SR Animal Sciences
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11th grade

Abstract

Dogs truly are man's best friends—and science's. Although dogs unfortunately age more rapidly than humans, they have a rich genetic make-up similar to humans, share our living environment, and face many of the same diseases we do. By studying health trajectories in dogs, we expand our knowledge of the aging process not just in our furry companions, but also in humans.

This is a cross-sectional study using data from The Dog Aging Project, and focuses on the prevalences of the most common health conditions in the most common dog breeds, and analyzes their correlation to different factors like sex, neuter status, size, and age.

There has been a 'replication crisis' in many scientific fields—groundbreaking results are emphasized over reproducibility and successful replication of said results. A famous example of the importance of replicability is how Nobel laureate Linus Pauling's claim that vitamin C lessens the risk of common colds was disproved by many subsequent studies. Thus, the first part of this study takes advantage of the open-source quality of The Dog Aging Project, and is a replication of another study: "Lifetime prevalence of owner-reported medical conditions in the 25 most common dog breeds in the Dog Aging Project pack" by Forsyth et al. (2023). Then, this project utilizes Python's statistical computing power to perform original number-based analyses regarding these most common medical conditions.

Findings include mixed vs pure breeds being more susceptible to certain medical conditions, as well as factors like sex and neuter status being found significant in predicting these conditions.

This research seeks to bridge the fields of veterinary science, animal science, and human health through an interdisciplinary and statistically sound approach. We aim to advance knowledge on how health trajectories can inform better practices, foster innovations in both animal and human medicine, and remediate the current 'replication crisis' in applied statistics fields.

Introduction

Dogs truly are man's best friends—and science's.

- Dogs share our environment and habits in ways that specimens like laboratory mice *do not*
    - E.g. Alzheimer's needs to be *artificially* induced in laboratory mice for research, whereas dogs develop Alzheimer's *naturally* like humans
- Dogs have a rich **genetic make-up similar to humans** and **share our home living environments**
- The study of health trajectories in dogs has far-reaching implications for veterinary science, animal science, and human health
- By studying health trajectories in dogs, we expand our knowledge of **aging** not just in our furry friends, but also in **humans too**

This project is a cross-sectional study using data from **The Dog Aging Project.**

- The Dog Aging Project is a 10-year longitudinal study sponsored by the **NIH** to understand how genes, lifestyle, and environment influence health and aging
- This paper focuses on the **prevalences** of the most common health conditions in common dog breeds
    - We analyze their **correlation** to different factors like sex, neuter status, size, and age

Introduction (cont.)

The purpose of this study addresses <u>two</u> main areas:

1) ADVANCING ANIMAL SCIENCE/VETERINARY MEDICINE

- Veterinary medicine, unlike human medicine, is generally **not personalized** — broad treatment protocols with a one-size-fits-all approach
- General wellness exams exclude breed-specific/age-specific testing unless symptoms have already appeared → leads to **late diagnoses,** limiting treatment options
- Knowing the prevalence of medical conditions based on breed/size/age improves **early detection**

This project advances a more **proactive, data-driven approach** to veterinary care, advocating for longer, healthier lives for pets.

2) ADVANCING STATISTICAL INTEGRITY

- Current **replication crisis** in applied statistics
  - Findings across scientific fields *fail* to hold up under replication and reproduction
- Undermines trust and leads to **wasted resources** when policies are built on unreliable data
- **Historical example** of a claim that has been disproved through statistical replication and verification:
  - Vitamin C doesn't actually prevent or cure the common cold

This project addresses this problem by **verifying** findings, identifying errors, and **refining** statistical methods, **uncovering** information deeply relevant to veterinary care with **statistical integrity.**

Methods

The following steps explain our procedure for this study in a manner such that others can replicate the general framework of our process: (more detailed procedure outlined in engineering notebook)

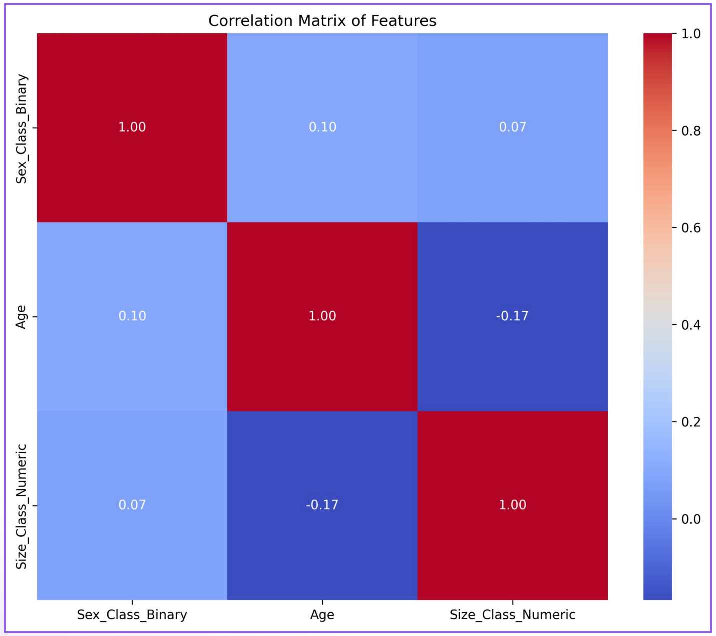
1. Obtain access to data (csv files) from The Dog Aging Project via their website.
2. Create a Jupyter notebook and import the datasets.
3. Perform **exploratory data analysis** on the chosen datasets to better understand the data in hand.
4. Read the chosen csv fields and clean them (removing rows with null values, removing columns that will not be used in the analysis, etc.).
5. Calculate the prevalences of the medical conditions being examined.
6. Perform **logistic regression** to obtain log-odds and p-values to interpret results.
    a. Choose the **predictor variable(s)** and the **dependent/response variable.**
    b. For this study, the dependent/response variable was whether there is presence of a certain medical condition or not, and the predictor variables included breed class (mixed/pure), size, age, sex, and sex class (neutered/intact).
7. Identify **statistically significant/discernible results** based on the p-values obtained.
    a. This study used the traditional $p<0.05$ as a threshold for 'significance,' but nuances are discussed in the Results & Discussion section of our engineering notebook.
8. For a replication study, use metrics like Jaccard index/error, precision, recall, F1 score, etc. to **measure similarities/differences.**

Methods (cont.)

$$Ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k$$

We used a **binomial model with logit link**; equation as above.

- Logistic regression maps predictions to the [0,1] range, making it ideal for classification
- Is more robust to skewed data



We used data visualization methods like correlation matrices to **discern feature relationships** and **detect multicollinearity**.

- Threshold: correlation<0.7
- Ensuring results are stable and interpretable

```
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```
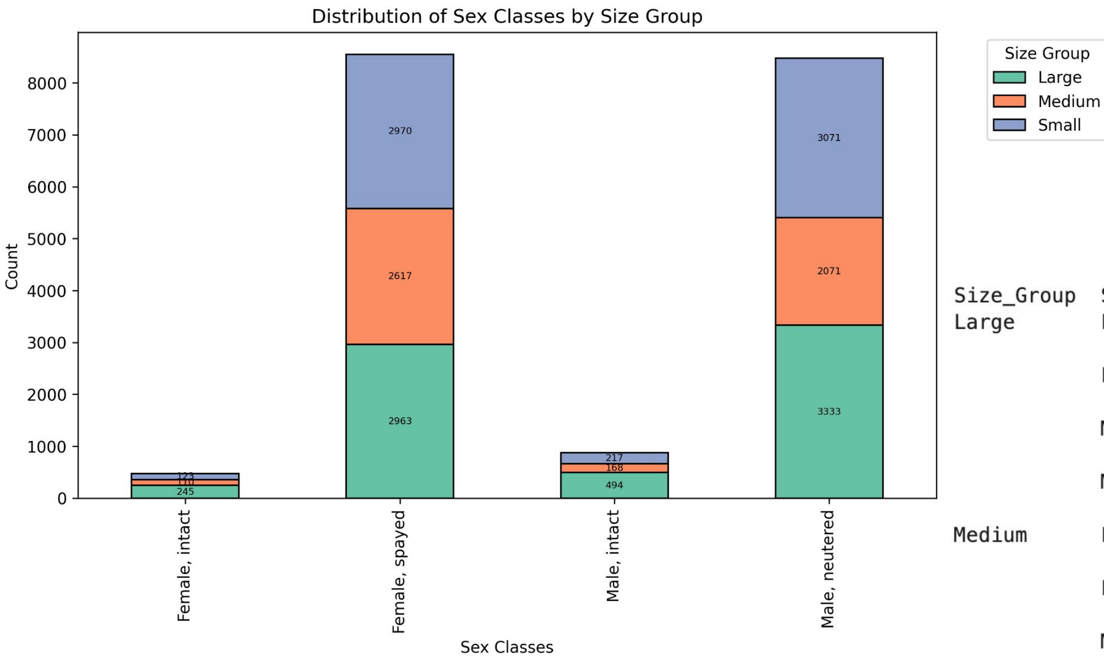

We used mle_retvals (Maximum Likelihood Estimation return values) to analyze and interpret the fitted model.

- It tells us our model **converged successfully**

Methods (cont.)



Our dataset was very imbalanced in sex class

- Far more neutered/spayed dogs than intact dogs
- Became a hurdle when investigating the impact of neuter status on health condition prevalences

```
Size_Group  Sex_Class_at_HLES  Breed_Class
Large       Female, intact     AKC-Recognized Breed                     16
                               Non-AKC-Recognized or Mixed Breed        16
            Female, spayed     AKC-Recognized Breed                     32
                               Non-AKC-Recognized or Mixed Breed        32
            Male, intact       AKC-Recognized Breed                     45
                               Non-AKC-Recognized or Mixed Breed        45
            Male, neutered     AKC-Recognized Breed                     90
                               Non-AKC-Recognized or Mixed Breed        90
Medium      Female, intact     AKC-Recognized Breed                     34
                               Non-AKC-Recognized or Mixed Breed        34
            Female, spayed     AKC-Recognized Breed                     68
                               Non-AKC-Recognized or Mixed Breed        68
            Male, intact       AKC-Recognized Breed                     36
                               Non-AKC-Recognized or Mixed Breed        36
            Male, neutered     AKC-Recognized Breed                     72
                               Non-AKC-Recognized or Mixed Breed        72
Small       Female, intact     AKC-Recognized Breed                     35
                               Non-AKC-Recognized or Mixed Breed        35
            Female, spayed     AKC-Recognized Breed                     70
                               Non-AKC-Recognized or Mixed Breed        70
            Male, intact       AKC-Recognized Breed                     59
                               Non-AKC-Recognized or Mixed Breed        59
            Male, neutered     AKC-Recognized Breed                    118
                               Non-AKC-Recognized or Mixed Breed       118
dtype: int64
```

To solve this, we downsampled to ensure the sample would be balanced

- Kept the amount of mixed breed/purebred intact/neutered males/females even
- The logit link model is more robust against other data imbalances

Results

Mixed breed vs. purebred prevalence table by medical condition

- (Image only contains half of the full table due to powerpoint size limitations. Full table in engineering notebook.)
- **Statistically significant/ discernible** results ($p<0.05$) obtained through binomial model with logit link are bolded
- 25 condition overlap

	N (mixed-breed/purebred)	Non-AKC-Recognized or Mixed Breed Prevalence (%)	AKC-Recognized Breed Prevalence (%)	p-value
Conjunctivitis	398/361	2.797498	2.728028	0.79
Extracted teeth	1901/1899	13.361917	14.350487	**1.36e-03**
Entropion (eyelid rolled in)	42/131	0.295213	0.989949	**5.11e-12**
Urinary incontinence	310/324	2.178956	2.448424	0.09
Torn or broken toenail	813/630	5.714487	4.760825	**1.00e-03**
Retained deciduous (baby) teeth	98/145	0.688831	1.095745	**3.18e-04**
Hip dysplasia	312/325	2.193013	2.455981	0.10
Hypothyroidism (low thyroid function)	254/266	1.785338	2.010126	0.12
Coccidia	184/251	1.293316	1.896773	**6.46e-05**
Lyme disease	386/323	2.713151	2.440868	0.23
Stenotic/narrow nares	5/51	0.035144	0.385400	**2.22e-06**
Food or medicine allergies	318/326	2.235187	2.463538	0.16
Fractured teeth	1002/776	7.042947	5.864128	**2.43e-04**
Lick granuloma	148/138	1.040276	1.042847	0.87
Hookworms	396/308	2.783440	2.327515	**1.81e-02**
Cruciate ligament rupture	566/414	3.978351	3.128542	**3.73e-04**
Chocolate	757/533	5.320869	4.027809	**1.47e-06**
Urinary crystals or stones in bladder or urethra	195/227	1.370633	1.715408	**1.23e-02**
Sebaceous cysts	469/473	3.296549	3.574397	0.12
Gingivitis (red, puffy gums)	416/471	2.924018	3.559284	**1.06e-03**
Chronic or recurrent hot spots	390/436	2.741267	3.294793	**3.26e-03**
Hearing loss (incompletely deaf)	311/342	2.185984	2.584448	**1.73e-02**
Ear Infection	869/1120	6.108104	8.463689	**1.73e-15**
Cataracts	535/617	3.760455	4.662586	**2.92e-05**

- Certain conditions are **more prevalent in either mixed or purebred dogs**
  - E.g. — Ear infections and stenotic/narrow nares are more statistically prevalent in purebred dogs, whereas fractured teeth is more statistically prevalent in mixed breed dogs.

Results (cont.)

Logistic regression results investigating sex class (neuter status), size class, and age as variables in the prevalence of health conditions. (Discussed on next slide.)

Term	Intercept				Sex_Class_Binary				Size_Class_Numeric				Age				Sex_Size				Sex_Age				Size_Age			
	CI_lower	CI_upper	Coefficient	P-value	CI_lower	CI_upper	Coefficient	P-value	CI_lower	CI_upper	Coefficient	P-value	CI_lower	CI_upper	Coefficient	P-value	CI_lower	CI_upper	Coefficient	P-value	CI_lower	CI_upper	Coefficient	P-value	CI_lower	CI_upper	Coefficient	P-value
Health_Condition																												
Anal sac impaction	-3.52811	-2.84773	**-3.1879**	0	-1.12766	1.102867	-0.0124	0.9826	-1.00344	0.939734	-0.0319	0.9488	-1.25633	0.861688	-0.1973	0.7150	-1.32699	1.038328	-0.1443	0.8110	-0.76946	1.097198	0.1639	0.7308	-0.79076	1.170959	0.1901	0.7040
Bordetella and/or parainfluenza (kennel cough)	-3.28098	-2.61797	**-2.9495**	0	-0.25767	1.819441	0.7809	0.1406	-0.14585	1.526013	0.6901	0.1057	-1.73059	0.503843	-0.6134	0.2819	-1.91761	0.165016	-0.8763	0.0991	-0.77737	0.991676	0.1072	0.8123	-0.55679	1.144567	0.2939	0.4983
Cataracts	-4.34225	-3.15133	**-3.7468**	0	-1.21268	1.713974	0.2506	0.7371	-2.30358	1.08702	-0.6083	0.4819	0.407479	2.946823	**1.6772**	**0.0096**	-1.12968	1.362804	0.1166	0.8546	-1.50671	0.846577	-0.3301	0.5825	-1.13889	1.584227	0.2227	0.7486
Chocolate	-3.05538	-2.45159	**-2.7535**	0	0.566532	2.831931	**1.6992**	**0.0033**	0.100904	2.014681	**1.0578**	**0.0303**	-0.38668	1.677078	0.6452	0.2204	-2.40316	-0.30158	**-1.3524**	**0.0117**	-1.52967	0.164435	-0.6826	0.1142	-0.91788	0.626775	-0.1456	0.7118
Chronic or recurrent diarrhea	-3.38806	-2.71259	**-3.0503**	0	-1.24565	1.055638	-0.0950	0.8714	-0.11903	1.527587	0.7043	0.0936	-0.60969	1.413418	0.4019	0.4362	-1.12714	1.131779	0.0023	0.9968	-0.61746	1.271988	0.3273	0.4972	-2.12888	-0.26204	**-1.1955**	**0.0121**
Chronic or recurrent hot spots	-3.78525	-2.95437	**-3.3698**	0	-1.08517	1.887757	0.4013	0.5967	-0.97077	1.583766	0.3065	0.6381	-0.88809	1.640469	0.3762	0.5598	-1.2926	1.403629	0.0555	0.9357	-1.219	0.983269	-0.1179	0.8338	-0.79879	1.181965	0.1916	0.7046
Coccidia	-4.52185	-3.37018	**-3.946**	0	-1.1011	1.706819	0.3029	0.6724	-0.48303	1.561982	0.5395	0.3011	-3.24079	0.461492	-1.3896	0.1412	-2.16548	0.782006	-0.6917	0.3576	-0.64133	2.330194	0.8444	0.2653	-1.28516	1.495787	0.1053	0.8820
Conjunctivitis	-3.75057	-2.98107	**-3.3658**	0	-1.20674	0.946155	-0.1303	0.8125	-1.15192	0.690971	-0.2305	0.6240	-1.8751	0.459064	-0.7080	0.2344	-1.11633	1.306696	0.0952	0.8776	-0.94243	1.188603	0.1231	0.8209	-0.86795	1.321407	0.2267	0.6848
Corneal ulcer	-4.84311	-3.60094	**-4.222**	0	-2.96898	0.639878	-1.1646	0.2059	-2.36297	1.127808	-0.6176	0.4880	-1.20869	1.641049	0.2162	0.7662	-1.00068	2.651719	0.8255	0.3756	-0.58325	2.109085	0.7629	0.2667	-1.49167	1.722798	0.1156	0.8879
Cruciate ligament rupture	-4.40079	-3.29589	**-3.8483**	0	-1.66858	1.907466	0.1194	0.8958	-0.22566	2.633804	1.2041	0.0988	-0.52534	2.483916	0.9793	0.2021	-1.98706	0.94595	-0.5206	0.4866	-0.78306	1.499433	0.3582	0.5385	-1.60875	0.810753	-0.3990	0.5180
Deafness	-4.80246	-3.51865	**-4.1606**	0	-4.03151	-0.68352	**-2.3575**	**0.0058**	-2.10225	1.056013	-0.5231	0.5162	-0.24711	2.066914	0.9099	0.1232	-0.19595	3.124269	1.4642	0.0839	0.442868	2.754391	**1.5986**	**0.0067**	-2.26575	0.73739	-0.7642	0.3185
Dental calculus (yellow build-up on teeth)	-1.62889	-1.23255	**-1.4307**	0	-0.70094	0.411253	-0.1448	0.6097	-1.80673	-0.51718	**-1.162**	**0.0004**	-0.09141	0.976636	0.4426	0.1043	-0.2144	1.077077	0.4313	0.1905	-0.80619	0.18959	-0.3083	0.2249	0.107643	1.253204	**0.6804**	**0.0199**
Dog bite	-2.09388	-1.68794	**-1.8909**	0	-0.45634	1.023921	0.2838	0.4523	0.086461	1.287563	**0.687**	**0.025**	0.298834	1.536247	**0.9175**	**0.0037**	-1.01277	0.370319	-0.3212	0.3626	-0.81151	0.290086	-0.2607	0.3536	-1.01826	0.081215	-0.4685	0.0948
Ear Infection	-2.7449	-2.20926	**-2.4771**	0	-0.73866	1.038052	0.1497	0.7412	-0.18649	1.22281	0.5182	0.1495	-1.47449	0.276742	-0.5989	0.1801	-1.20011	0.552694	-0.3237	0.4691	-0.21368	1.163554	0.4749	0.1764	-0.335	1.046968	0.3560	0.3126
Extracted teeth	-1.93647	-1.47195	**-1.7042**	0	-0.62503	0.546877	-0.0391	0.8960	-1.65221	-0.25972	**-0.956**	**0.0071**	-0.38562	0.780866	0.1976	0.5066	-0.92636	0.501996	-0.2122	0.5604	-0.31133	0.711669	0.2002	0.4431	-0.0583	1.247273	0.5945	0.0743
Fleas	-3.34301	-2.70266	**-3.0228**	0	-0.97722	1.004802	0.0138	0.9782	-1.28486	0.608571	-0.3381	0.4839	-1.34901	0.598726	-0.3751	0.4503	-1.26514	0.95094	-0.1571	0.7811	-0.69905	1.013672	0.1573	0.7188	-0.39701	1.454745	0.5289	0.2629
Food or medicine allergies	-3.78303	-3.00148	**-3.3923**	0	-1.40211	1.033751	-0.1842	0.7669	-1.22291	0.872874	-0.1750	0.7434	-2.06141	0.443051	-0.8092	0.2053	-1.11532	1.539372	0.2120	0.7542	-0.61883	1.576549	0.4789	0.3925	-0.66338	1.437806	0.3872	0.4701
Food or medicine allergies that affect the skin	-4.07078	-3.19817	**-3.6345**	0	-1.2041	1.537797	0.1668	0.8115	-1.60792	1.046434	-0.2807	0.6784	-0.82025	1.550571	0.3652	0.5460	-1.10074	1.699221	0.2992	0.6753	-2.0228	0.317884	-0.8525	0.1534	-0.72057	1.635201	0.4573	0.4467
Foreign body ingestion or blockage	-4.04691	-3.17569	**-3.6113**	0	-0.96804	2.277073	0.6545	0.4292	-0.35028	2.171624	0.9107	0.1569	-0.06723	2.524621	1.2287	0.0631	-1.60623	1.402167	-0.1020	0.8942	-2.07611	0.387436	-0.8443	0.1791	-2.21092	0.055566	-1.0777	0.0623
Fractured teeth	-2.90545	-2.35831	**-2.6319**	0	-0.44135	1.376495	0.4676	0.3133	-0.35134	1.198091	0.4234	0.2841	-0.6083	1.069349	0.2305	0.5901	-1.69521	0.080901	-0.8072	0.0748	-0.91168	0.510536	-0.2006	0.5804	-0.4496	1.047054	0.2987	0.4340
Giardia	-2.56442	-2.08028	**-2.3224**	0	-0.96821	0.535211	-0.2165	0.5724	-0.22873	0.911999	0.3416	0.2404	-1.3724	0.174091	-0.5992	0.1288	-0.89458	0.657948	-0.1183	0.7651	-0.14188	1.191868	0.5250	0.1228	-0.79121	0.558559	-0.1163	0.7355
Gingivitis (red, puffy gums)	-4.12742	-3.05911	**-3.5933**	0	-0.70553	1.763708	0.5291	0.4009	-2.35163	0.742444	-0.8046	0.3080	-0.94186	1.469473	0.2638	0.6680	-2.03015	0.802693	-0.6137	0.3957	-1.14081	0.90087	-0.1200	0.8178	-0.44857	2.181757	0.8666	0.1965
Hearing loss (incompletely deaf)	-5.84035	-3.89163	**-4.866**	0	-0.93583	4.42609	1.7451	0.2020	-1.76589	3.071772	0.6529	0.5968	0.362106	4.817241	**2.5897**	**0.0227**	-2.63415	0.653823	-0.9902	0.2378	-2.72151	1.14129	-0.7901	0.4227	-1.77748	1.808337	0.0154	0.9865
Hip dysplasia	-4.61821	-3.44859	**-4.0334**	0	-1.77094	2.302073	0.2656	0.7983	-0.1516	2.916534	1.3825	0.0773	-0.36316	2.859188	1.2480	0.1290	-2.12227	1.27591	-0.4232	0.6254	-1.32303	1.270313	-0.0264	0.9682	-2.08031	0.509091	-0.7856	0.2343
Hookworms	-3.99213	-3.13932	**-3.5657**	0	-0.99393	1.695931	0.3510	0.6090	-0.48955	1.604531	0.5575	0.2967	-1.12993	1.440811	0.1554	0.8126	-1.59926	1.027473	-0.2859	0.6696	-1.60883	0.702717	-0.4531	0.4423	-1.38032	0.890969	-0.2447	0.6728
Hypothyroidism (low thyroid function)	-4.85869	-3.53171	**-4.1952**	0	-0.80661	3.323753	1.2586	0.2323	-2.25532	1.665024	-0.2951	0.7679	-1.01158	2.550306	0.7694	0.3972	-1.85604	1.620621	-0.1177	0.8944	-2.8727	0.355605	-1.2585	0.1265	-0.40486	2.495248	1.0452	0.1577
Intervertebral disc disease (IVDD) (Orthopedic)	-6.39553	-3.8799	**-5.1377**	0	-2.37461	2.898058	0.2617	0.8457	-5.26723	2.521761	-1.3727	0.4897	-0.67772	3.466514	1.3944	0.1872	-2.74862	4.288182	0.7698	0.6681	-2.52105	1.364985	-0.5780	0.5598	-2.82009	2.675353	-0.0724	0.9588
Keratoconjunctivitis sicca (KCS)	-4.88544	-3.54759	**-4.2165**	0	-0.46031	3.639467	1.5896	0.1285	-2.2202	1.883219	-0.1685	0.8721	-1.11854	2.597785	0.7396	0.4353	-2.5784	1.080136	-0.7491	0.4222	-2.71465	0.518939	-1.0979	0.1832	-0.60088	2.462313	0.9307	0.2336
Laceration	-4.9414	-3.55563	**-4.2485**	0	-0.33692	3.454519	1.5588	0.1070	0.516324	3.239494	**1.8779**	**0.0069**	-0.10393	3.096872	1.4965	0.0668	-2.99977	0.364126	-1.3178	0.1246	-3.46589	-0.48544	**-1.9757**	**0.0094**	-2.21985	0.418205	-0.9008	0.1807
Lick granuloma	-5.58491	-3.79974	**-4.6923**	0	-1.95194	3.445468	0.7468	0.5876	-3.07728	1.929841	-0.5737	0.6533	-0.46931	3.608013	1.5694	0.1314	-1.03758	3.371737	1.1671	0.2995	-3.67281	0.522539	-1.5751	0.1411	-1.36019	2.160378	0.4001	0.6560
Lyme disease	-3.44159	-2.73179	**-3.0867**	0	0.257292	2.865833	**1.5616**	**0.0189**	-0.326	1.922732	0.7984	0.1640	-0.9473	1.490733	0.2717	0.6622	-2.43747	0.058443	-1.1895	0.0617	-1.44054	0.575789	-0.4324	0.4006	-0.80216	0.971628	0.0847	0.8515
Murmur	-3.22803	-2.50966	**-2.8688**	0	-0.64545	1.46008	0.4073	0.4483	-1.55576	0.710607	-0.4226	0.4648	-0.38843	1.501414	0.5565	0.2484	-1.30047	0.816878	-0.2418	0.6544	-0.78954	0.897966	0.0542	0.8998	-0.50771	1.314894	0.4036	0.3854
Osteoarthritis	-4.31955	-3.11365	**-3.7166**	0	-1.57863	2.213599	0.3175	0.7428	-0.91956	2.139852	0.6101	0.4344	0.021404	3.011702	**1.5166**	**0.0468**	-1.11153	1.482537	0.1855	0.7792	-1.29002	1.378456	0.0442	0.9482	-1.22093	1.024699	-0.0981	0.8640
Pancreatitis	-3.98748	-3.14606	**-3.5668**	0	-1.19763	1.881699	0.3420	0.6633	-1.32744	1.482452	0.0775	0.9139	-0.75141	1.832501	0.5405	0.4122	-1.59315	1.62538	0.0161	0.9843	-1.29707	1.093363	-0.1019	0.8674	-1.54495	0.780015	-0.3825	0.5190
Patellar luxation	-4.23096	-3.0048	**-3.6179**	0	-0.20019	2.077459	0.9386	0.1062	-2.24092	0.709143	-0.7659	0.3088	-0.5081	1.727428	0.6097	0.2851	-2.85636	0.885764	-0.9853	0.3020	-1.71058	1.31E-05	-0.8553	0.0500	-1.73897	1.354858	-0.1921	0.8077
Pruritis (itchy skin)	-3.0915	-2.49739	**-2.7944**	0	-1.33423	0.623807	-0.3552	0.4770	-1.12735	0.605948	-0.2607	0.5555	-0.60963	1.049577	0.2200	0.6033	-0.47991	1.718627	0.6194	0.2695	-0.62008	1.010355	0.1951	0.6390	-1.50427	0.21364	-0.6453	0.1409
Retained deciduous (baby) teeth	-8.36779	-3.17764	**-5.7727**	0	-1.66703	2.052575	0.1928	0.8390	-4.34862	1.273348	-1.5376	0.2837	-1.93742	3.89825	0.9804	0.5102	-2.69695	4.592338	0.9477	0.6103	-2.25342	0.71085	-0.7713	0.3078	-8.75389	3.006136	-2.8739	0.3381
Roundworms	-3.58913	-2.88638	**-3.2378**	0	-0.85538	1.538055	0.3413	0.5761	-0.52276	1.406723	0.4420	0.3692	-1.25068	1.076591	-0.0870	0.8834	-1.75001	0.688963	-0.5305	0.3938	-0.81535	1.171603	0.1781	0.7253	-1.17505	0.828473	-0.1733	0.7346
Seasonal allergies	-2.71555	-2.20743	**-2.4615**	0	-0.56546	1.180317	0.3074	0.4900	-0.68122	0.872449	0.0956	0.8094	-0.71432	0.868912	0.0773	0.8482	-0.90659	0.811834	-0.0474	0.9139	-1.01879	0.384145	-0.3173	0.3753	-0.27751	1.103798	0.4131	0.2410
Sebaceous cysts	-3.74803	-2.91564	**-3.3318**	0	-0.47966	2.077617	0.7990	0.2207	-0.59036	1.754233	0.5819	0.3306	-0.41985	1.932762	0.7565	0.2075	-2.54403	-0.1235	**-1.3338**	**0.0308**	-1.02198	0.878222	-0.0719	0.8821	-0.94174	1.172904	0.1156	0.8303
Seizures (including epilepsy)	-4.52603	-3.39552	**-3.9608**	0	-2.86242	0.140679	-1.3609	0.0757	-2.3007	0.665991	-0.8174	0.2802	-1.10721	1.279164	0.0860	0.8877	-0.67212	2.329403	0.8286	0.2792	-0.37434	1.901992	0.7638	0.1884	-1.02455	1.81416	0.3948	0.5856
Torn or broken toenail	-3.06503	-2.48316	**-2.7741**	0	-0.46886	1.432073	0.4816	0.3206	-0.73787	0.96302	0.1126	0.7953	-1.09147	0.749961	-0.1708	0.7162	-1.18163	0.759152	-0.2112	0.6696	-1.24052	0.372089	-0.4342	0.2912	-0.29138	1.288341	0.4985	0.2161
Tracheal collapse	-7685.62	7647.369	-19.1253	0.9961	-8487.4	8487.04	-0.1817	1.0000	-17659.8	17634.25	-12.7607	0.9989	-9050.85	9057.271	3.2107	0.9994	-22650.9	22653.67	1.3949	0.9999	-2.60757	2.035349	-0.2861	0.8091	-19983.6	19975.09	-4.4228	0.9997
Underbite	-4.68037	-3.52802	**-4.1042**	0	-2.07867	0.898419	-0.5901	0.4372	-2.3972	0.577742	-0.9097	0.2306	-2.36537	0.731669	-0.8169	0.3012	-1.20646	2.745761	0.7696	0.4452	-0.93268	2.046714	0.5570	0.4636	-1.65392	1.644692	-0.0046	0.9956
Urinary crystals or stones in bladder or urethra	-4.26386	-3.31327	**-3.7886**	0	-1.35553	2.017787	0.3311	0.7004	-0.75348	2.001814	0.6242	0.3745	-0.02048	2.573724	1.2766	0.0537	-1.78381	1.267478	-0.2582	0.7401	-1.88898	0.560476	-0.6643	0.2878	-1.69641	0.714874	-0.4908	0.4250
Urinary incontinence	-4.52959	-3.44072	**-3.9852**	0	-1.72213	1.961443	0.1197	0.8987	-0.46445	2.321874	0.9287	0.1914	-0.57381	2.381066	0.9036	0.2306	-2.01165	1.167915	-0.4219	0.6030	-1.5	1.027988	-0.2360	0.7144	-1.58528	0.982808	-0.3012	0.6457

Discussion

Jaccard Index: Measures the proportion of overlap relative to the total unique items:

$$Jaccard\ Index = \frac{Shared\ Items}{Union} = \frac{25}{33} \approx 0.7576$$

Jaccard Error: Measures the proportion of non-overlapping items:

$$Jaccard\ Error = 1 - Jaccard\ Index = 1 - 0.7576 \approx 0.2424$$

- Our results for the prevalence of health conditions in mixed vs purebred dogs were **similar** to Forsyth et al. (2023), with 25 overlapping conditions where breed class was found significant
    - 5 were found significant in our results and not Forsyth et al. (2023), and 3 were found significant in Forsyth et al. (2023) but not ours
    - We used numerical methods to quantify similarities and differences, like jaccard index and jaccard error (in the image above)
    - Discrepancies could be due to the fact that our starting datasets were not identical
        - Forsyth et al. (2023) used an **earlier version** of The Dog Aging Project's data release that is no longer available
- In our original analysis, we find that certain combinations of factors serve as relevant predictors for certain health conditions
    - We investigated the 7 interactions between sex class (neuter status), size class, and age
    - Results should be discussed with domain knowledge to be contextualized
- We also used additional statistical methods like lasso regression to regularize our variable selection and provide more insight into our regression results

Conclusion

- Mixed breeds were found to be more/less **predisposed** to certain health conditions, depending on the condition
  - Key insight that will enhance the quality and ability of veterinary medicine
- Certain factors proved to be more relevant in predicting different health conditions
  - Opens pathways for veterinary medicine to become **more personalized**
  - Enables more **early detection** for common health conditions
  - Allows for a step away from one-size-fits-all approaches towards healthcare
- This study backed the **statistical integrity** and foundation of other studies in this field, specifically Forsyth et al. (2023)
  - Working towards **addressing the replication crisis** in applied statistics
  - Advancing stronger **numerical integrity** within the statistics field
- This type of data analysis also holds potential in researching **human health**

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